O*NET OnLine

Computer Systems Analysts

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Bright Outlook | Updated 2026

Analyze science, engineering, business, and other data processing problems to develop and implement solutions to complex applications problems, system administration issues, or network concerns. Perform systems management and integration functions, improve existing computer systems, and review computer system capabilities, workflow, and schedule limitations. May analyze or recommend commercially available software.

Sample of reported job titles: Applications Analyst, Business Systems Analyst, Computer Analyst, Computer Systems Analyst, Computer Systems Consultant, Information Systems Analyst (ISA), IT Analyst (Information Technology Analyst), IT Systems Analyst (Information Technology Systems Analyst), Programmer Analyst, Systems Analyst

Also see: Health Informatics Specialists

Summary | Details | Custom | Easy Read | Veterans | Español

Contents

Occupation-Specific Information

Tasks

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Importance	Category	Task
81	Core	Troubleshoot program and system malfunctions to restore normal functioning.
76	Core	Provide staff and users with assistance solving computer-related problems, such as malfunctions and program problems.

Importance	Category	Task
75	Core	Test, maintain, and monitor computer programs and systems, including coordinating the installation of computer programs and systems.
74	Core	Use the computer in the analysis and solution of business problems, such as development of integrated production and inventory control and cost analysis systems.
70	Core	Coordinate and link the computer systems within an organization to increase compatibility so that information can be shared.
68	Core	Use object-oriented programming languages, as well as client and server applications development processes and multimedia and Internet technology.
65	Core	Analyze information processing or computation needs and plan and design computer systems, using techniques such as structured analysis, data modeling, and information engineering.
64	Core	Consult with management to ensure agreement on system principles.
62	Core	Specify inputs accessed by the system and plan the distribution and use of the results.
62	Core	Expand or modify system to serve new purposes or improve work flow.
61	Core	Train staff and users to work with computer systems and programs.
59	Core	Assess the usefulness of pre-developed application packages and adapt them to a user environment.
58	Core	Determine computer software or hardware needed to set up or alter systems.
55	Core	Read manuals, periodicals, and technical reports to learn how to develop programs that meet staff and user requirements.
55	Core	Develop, document, and revise system design procedures, test procedures, and quality standards.

Importance	Category	Task
53	Core	Recommend new equipment or software packages.
52	Core	Define the goals of the system and devise flow charts and diagrams describing logical operational steps of programs.
52	Core	Confer with clients regarding the nature of the information processing or computation needs a computer program is to address.
65	Supplemental	Review and analyze computer printouts and performance indicators to locate code problems, and correct errors by correcting codes.
61	Supplemental	Interview or survey workers, observe job performance, or perform the job to determine what information is processed and how it is processed.
60	Supplemental	Supervise computer programmers or other systems analysts or serve as project leaders for particular systems projects.
45	Supplemental	Prepare cost-benefit and return-on-investment analyses to aid in decisions on system implementation.
Not available	New	Write code to perform desired actions.

Find occupations related to multiple tasks

Occupational Requirements

Work Activities

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Importance	Work Activity
99	**Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.

Importance	Work Activity
90	**Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
90	**Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.
86	**Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.
86	**Updating and Using Relevant Knowledge** — Keeping up-to-date technically and applying new knowledge to your job.
79	**Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.
79	**Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.
76	**Monitoring Processes, Materials, or Surroundings** — Monitoring and reviewing information from materials, events, or the environment, to detect or assess problems.
70	**Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.
67	**Evaluating Information to Determine Compliance with Standards** — Using relevant information and individual judgment to determine whether events or processes comply with laws, regulations, or standards.
67	**Interpreting the Meaning of Information for Others** — Translating or explaining what information means and how it can be used.
66	**Judging the Qualities of Objects, Services, or People** — Assessing the value, importance, or quality of things or people.
66	**Organizing, Planning, and Prioritizing Work** — Developing specific goals and plans to prioritize, organize, and accomplish your work.
66	**Thinking Creatively** — Developing, designing, or creating new applications, ideas, relationships, systems, or products, including artistic contributions.

Importance	Work Activity
65	**Documenting/Recording Information** — Entering, transcribing, recording, storing, or maintaining information in written or electronic/magnetic form.
64	**Establishing and Maintaining Interpersonal Relationships** — Developing constructive and cooperative working relationships with others, and maintaining them over time.
62	**Developing Objectives and Strategies** — Establishing long-range objectives and specifying the strategies and actions to achieve them.
60	**Inspecting Equipment, Structures, or Materials** — Inspecting equipment, structures, or materials to identify the cause of errors or other problems or defects.
57	**Assisting and Caring for Others** — Providing personal assistance, medical attention, emotional support, or other personal care to others such as coworkers, customers, or patients.
57	**Communicating with People Outside the Organization** — Communicating with people outside the organization, representing the organization to customers, the public, government, and other external sources. This information can be exchanged in person, in writing, or by telephone or e-mail.
57	**Controlling Machines and Processes** — Using either control mechanisms or direct physical activity to operate machines or processes (not including computers or vehicles).
55	**Estimating the Quantifiable Characteristics of Products, Events, or Information** — Estimating sizes, distances, and quantities; or determining time, costs, resources, or materials needed to perform a work activity.
54	**Developing and Building Teams** — Encouraging and building mutual trust, respect, and cooperation among team members.
53	**Resolving Conflicts and Negotiating with Others** — Handling complaints, settling disputes, and resolving grievances and conflicts, or otherwise negotiating with others.
50	**Scheduling Work and Activities** — Scheduling events, programs, and activities, as well as the work of others.
45	**Training and Teaching Others** — Identifying the educational needs of others, developing formal educational or training programs or classes, and teaching or instructing others.

Importance	Work Activity
43	**Monitoring and Controlling Resources** — Monitoring and controlling resources and overseeing the spending of money.
41	**Repairing and Maintaining Electronic Equipment** — Servicing, repairing, calibrating, regulating, fine-tuning, or testing machines, devices, and equipment that operate primarily on the basis of electrical or electronic (not mechanical) principles.
39	**Guiding, Directing, and Motivating Subordinates** — Providing guidance and direction to subordinates, including setting performance standards and monitoring performance.
38	**Coaching and Developing Others** — Identifying the developmental needs of others and coaching, mentoring, or otherwise helping others to improve their knowledge or skills.
37	**Coordinating the Work and Activities of Others** — Getting members of a group to work together to accomplish tasks.
37	**Providing Consultation and Advice to Others** — Providing guidance and expert advice to management or other groups on technical, systems-, or process-related topics.
36	**Performing Administrative Activities** — Performing day-to-day administrative tasks such as maintaining information files and processing paperwork.
31	**Handling and Moving Objects** — Using hands and arms in handling, installing, positioning, and moving materials, and manipulating things.
25	**Performing General Physical Activities** — Performing general physical activities includes doing activities that require considerable use of your arms and legs and moving your whole body, such as climbing, lifting, balancing, walking, stooping, and handling materials.
22	**Repairing and Maintaining Mechanical Equipment** — Servicing, repairing, adjusting, and testing machines, devices, moving parts, and equipment that operate primarily on the basis of mechanical (not electronic) principles.
20	**Drafting, Laying Out, and Specifying Technical Devices, Parts, and Equipment** — Providing documentation, detailed instructions, drawings, or specifications to tell others about how devices, parts, equipment, or structures are to be fabricated, constructed, assembled, modified, maintained, or used.

Importance	Work Activity
19	**Operating Vehicles, Mechanized Devices, or Equipment** — Running, maneuvering, navigating, or driving vehicles or mechanized equipment, such as forklifts, passenger vehicles, aircraft, or watercraft.
14	**Selling or Influencing Others** — Convincing others to buy merchandise/goods or to otherwise change their minds or actions.
13	**Performing for or Working Directly with the Public** — Performing for people or dealing directly with the public. This includes serving customers in restaurants and stores, and receiving clients or guests.
13	**Staffing Organizational Units** — Recruiting, interviewing, selecting, hiring, and promoting employees in an organization.

Detailed Work Activities

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- Troubleshoot issues with computer applications or systems.
- Provide technical support for software maintenance or use.
- Coordinate software or hardware installation.
- Monitor computer system performance to ensure proper operation.
- Test software performance.
- Apply information technology to solve business or other applied problems.
- Configure computer networks.
- Write computer programming code.
- Analyze project data to determine specifications or requirements.
- Design integrated computer systems.
- Analyze data to identify or resolve operational problems.
- Collaborate with others to determine design specifications or details.
- Modify software programs to improve performance.
- Select production input materials.
- Train others in computer interface or software use.
- Collect data about customer needs.
- Manage information technology projects or system activities.
- Supervise information technology personnel.
- Evaluate utility of software or hardware technologies.

- Identify information technology project resource requirements.
- Develop testing routines or procedures.
- Document design or development procedures.
- Read documents to gather technical information.
- Provide recommendations to others about computer hardware.
- Develop diagrams or flow charts of system operation.
- Estimate time or monetary resources needed to complete projects.

Find occupations related to multiple detailed work activities

Work Context

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- **E-Mail** — How frequently does your job require you to use E-mail?
  - 100% Every day
- **Telephone Conversations** — How often do you have telephone conversations in this job?
  - 79% Every day
  - 21% Once a week or more but not every day
- **Indoors, Environmentally Controlled** — How often does this job require working indoors in an environmentally controlled environment (like a warehouse with air conditioning)?
  - 93% Every day
- **Determine Tasks, Priorities and Goals** — How much freedom does the worker have in determining the tasks, priorities, or goals of the job?
  - 65% A lot of freedom
- **Importance of Being Exact or Accurate** — How important is being very exact or highly accurate in performing this job?
  - 54% Extremely important
  - 45% Very important
- **Contact With Others** — How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?
  - 73% Constant contact with others
  - 14% Contact with others most of the time
  - 11% Occasional contact with others
- **Work With or Contribute to a Work Group or Team** — How important is it to work with or contribute to a work group or team in this job?

63% Extremely important

22% Very important

14% Fairly important

- **Face-to-Face Discussions with Individuals and Within Teams** — How frequently does your job require face-to-face discussions with individuals and within teams?

  32% Every day

  63% Once a week or more but not every day

- **Freedom to Make Decisions** — How much decision making freedom, without supervision, does the job offer?

  34% A lot of freedom

  58% Some freedom

- **Spend Time Sitting** — How much does this job require sitting?

  53% Continually or almost continually

  22% More than half the time

  14% About half the time

  11% Less than half the time

- **Time Pressure** — How often does this job require the worker to meet strict deadlines?

  50% Once a week or more but not every day

  14% Once a month or more but not every week

- **Duration of Typical Work Week** — Number of hours typically worked in one week.

  56% More than 40 hours

  44% 40 hours

- **Importance of Repeating Same Tasks** — How important are continuous, repetitive, physical activities (like key entry) or mental activities (like checking entries in a ledger) to performing this job?

  45% Extremely important

  39% Very important

  12% Not important at all

- **Work Outcomes and Results of Other Workers** — How responsible is the worker for work outcomes and results of other workers?

  81% High responsibility

- **Impact of Decisions on Co-workers or Company Results** — What results do your decisions usually have on other people or the image or reputation or financial resources of your employer?

  61% Important results

  35% Moderate results

- **Spend Time Making Repetitive Motions** — How much does this job require making repetitive motions?

- 35% Continually or almost continually
- 28% More than half the time
- 33% Less than half the time

Coordinate or Lead Others in Accomplishing Work Activities — How important is it to coordinate or lead others (not as a supervisor or team leader) in accomplishing work activities in this job?

- 15% Extremely important
- 55% Very important
- 16% Important
- 15% Not important at all

Frequency of Decision Making — How often is the worker required to make decisions that affect other people, the financial resources, and/or the image and reputation of the organization?

- 15% Every day
- 23% Once a week or more but not every day
- 54% Once a month or more but not every week

Level of Competition — To what extent does this job require the worker to compete or to be aware of competitive pressures?

- 18% Extremely competitive
- 13% Highly competitive
- 61% Moderately competitive

Spend Time Using Your Hands to Handle, Control, or Feel Objects, Tools, or Controls — How much does this job require using your hands to handle, control, or feel objects, tools or controls?

- 42% Continually or almost continually
- 20% More than half the time
- 38% Never

Written Letters and Memos — How frequently does your job require written letters and memos?

- 15% Every day
- 29% Once a week or more but not every day
- 19% Once a month or more but not every week
- 34% Once a year or more but not every month

Consequence of Error — How serious would the result usually be if the worker made a mistake that was not easily correctable?

- 16% Very serious
- 43% Serious
- 32% Fairly serious

Physical Proximity — To what extent does this job require the worker to perform job tasks physically close to other people?

34% Moderately close (at arm's length)

32% Slightly close (e.g., shared office)

32% I work with others but not closely (e.g., private office)

Deal With External Customers or the Public in General — How important is it to deal with external customers (as in retail sales) or the public in general (as in police work) in this job?

44% Very important

12% Fairly important

34% Not important at all

Degree of Automation — How automated is the job?

17% Highly automated

43% Moderately automated

35% Slightly automated

Dealing With Unpleasant, Angry, or Discourteous People — How frequently does the worker have to deal with unpleasant, angry, or discourteous individuals as part of the job requirements?

14% Once a week or more but not every day

16% Once a month or more but not every week

66% Once a year or more but not every month

Spend Time Standing — How much does this job require standing?

14% More than half the time

17% About half the time

61% Less than half the time

Conflict Situations — How frequently are there conflict situations the employee has to face in this job?

21% Once a week or more but not every day

19% Once a month or more but not every week

34% Once a year or more but not every month

Public Speaking — How frequently does your job require public speaking (one speaker with an audience)?

11% Every day

19% Once a month or more but not every week

50% Once a year or more but not every month

20% Never

Spend Time Walking or Running — How much does this job require walking or running?

15% Continually or almost continually

52% Less than half the time

28% Never

- **Health and Safety of Other Workers** — How much responsibility is there for the health and safety of others in this job?
  - 14% High responsibility
  - 21% Moderate responsibility
  - 26% Limited responsibility
  - 38% No responsibility
- **Exposed to Sounds, Noise Levels that are Distracting or Uncomfortable** — How often does this job require working exposed to sounds and noise levels that are distracting or uncomfortable?
  - 30% Once a year or more but not every month
  - 55% Never
- **Wear Common Protective or Safety Equipment such as Safety Shoes, Glasses, Gloves, Hearing Protection, Hard Hats, or Life Jackets** — How often does this job require wearing common protective or safety equipment such as safety shoes, glasses, gloves, hearing protection, hard hats or life-jackets?
  - 76% Never
- **Exposed to Cramped Work Space, Awkward Positions** — How often does this job require working in cramped work spaces that requires getting into awkward positions?
  - 15% Once a month or more but not every week
  - 15% Once a year or more but not every month
  - 70% Never
- **In an Enclosed Vehicle or Operate Enclosed Equipment** — How often does this job require working in a closed vehicle or operate enclosed equipment (like a car)?
  - 11% Once a month or more but not every week
  - 15% Once a year or more but not every month
  - 72% Never
- **Spend Time Bending or Twisting Your Body** — How much does this job require bending or twisting your body?
  - 38% Less than half the time
  - 62% Never
- **Spend Time Kneeling, Crouching, Stooping, or Crawling** — How much does this job require kneeling, crouching, stooping or crawling?
  - 32% Less than half the time
  - 66% Never
- **Pace Determined by Speed of Equipment** — How important is it to this job that the pace is determined by the speed of equipment or machinery? (This does not refer to keeping busy at all times on this job.)
  - 31% Fairly important
  - 69% Not important at all

- **Exposed to Contaminants** — How often does this job require working exposed to contaminants (such as pollutants, gases, dust or odors)?
  - 18% Once a year or more but not every month
  - 79% Never
- **Exposed to Minor Burns, Cuts, Bites, or Stings** — How often does this job require exposure to minor burns, cuts, bites, or stings?
  - 12% Once a year or more but not every month
  - 85% Never
- **Dealing with Violent or Physically Aggressive People** — How frequently does this job require the worker to deal with physical aggression of violent individuals?
  - 17% Once a year or more but not every month
  - 83% Never
- **Exposed to Very Hot or Cold Temperatures** — How often does this job require working in very hot (above 90 F degrees) or very cold (below 32 F degrees) temperatures?
  - 88% Never
- **Indoors, Not Environmentally Controlled** — How often does this job require working in an environment that is not environmentally controlled (like a warehouse without air conditioning)?
  - 91% Never
- **Exposed to Disease or Infections** — How often does this job require exposure to disease/infections?
  - 93% Never
- **Exposed to Hazardous Conditions** — How often does this job require exposure to hazardous conditions?
  - 93% Never
- **Exposed to Hazardous Equipment** — How often does this job require exposure to hazardous equipment?
  - 11% Once a year or more but not every month
  - 89% Never
- **Exposed to High Places** — How often does this job require exposure to high places?
  - 12% Once a year or more but not every month
  - 88% Never
- **Outdoors, Exposed to All Weather Conditions** — How often does this job require working outdoors, exposed to all weather conditions?
  - 93% Never
- **Outdoors, Under Cover** — How often does this job require working outdoors, under cover (like in an open shed)?

94% Never

- **Spend Time Climbing Ladders, Scaffolds, or Poles** — How much does this job require climbing ladders, scaffolds, poles, etc.?

  12% Less than half the time

  88% Never

- **Exposed to Extremely Bright or Inadequate Lighting Conditions** — How often does this job require working in extremely bright or inadequate lighting conditions?

  93% Never

- **In an Open Vehicle or Operating Equipment** — How often does this job require working in an open vehicle or operating equipment (like a tractor)?

  95% Never

- **Spend Time Keeping or Regaining Balance** — How much does this job require keeping or regaining your balance?

  91% Never

- **Exposed to Radiation** — How often does this job require exposure to radiation?

  100% Never

- **Exposed to Whole Body Vibration** — How often does this job require exposure to whole body vibration (like operating a jackhammer or earth moving equipment)?

  100% Never

- **Wear Specialized Protective or Safety Equipment such as Breathing Apparatus, Safety Harness, Full Protection Suits, or Radiation Protection** — How often does this job require wearing specialized protective or safety equipment such as breathing apparatus, safety harness, full protection suits, or radiation protection?

  100% Never

- **Work Schedules** — How regular are the work schedules for this job?

  100% Regular (established routine, set schedule)

Experience Requirements

Job Zone

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Title	Job Zone Four: Considerable Preparation Needed
Education	Most of these occupations require a four-year bachelor's degree, but some do not.
Related Experience	A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of

	college and work for several years in accounting to be considered qualified.
Job Training	Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.
Job Zone Examples	Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, conservation scientists, art directors, and cost estimators.
SVP Range	(7.0 to < 8.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

State licenses Select a State Go

Apprenticeship Opportunities

All 5 displayed

Example apprenticeship titles for this occupation:

- Communications-Computer Systems Planning and Implementation(Mil Only)
- Computer Systems Analyst
- Computer Systems Analyst (Nof)
- It Analyst
- Public Health Informatician (Cdc)

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Access software** — Access management software; Citrix cloud computing software
- **Accounting software** — Fund accounting software; Tax software
- **Administration software** — Cisco Systems CiscoWorks; Element management software
- **Analytical or scientific software** — IBM SPSS Statistics; Minitab; SAS; The MathWorks MATLAB; 3 more
- **Application server software** — Docker; GitHub; Red Hat OpenShift; Spring Boot; 3 more
- **Backup or archival software** — System and data disaster recovery software; Veritas NetBackup
- **Business intelligence and data analysis software** — Apache Spark; IBM Cognos Impromptu; Microsoft Power BI; Tableau; 4 more
- **Cloud-based data access and sharing software** — Slack
- **Cloud-based management software** — Amazon Web Services AWS CloudFormation; IBM WebSphere; Oracle Cloud software; Splunk Enterprise
- **Clustering software** — VMware
- **Communications server software** — IBM Domino
- **Compiler and decompiler software** — Time sharing option TSO software
- **Computer aided design CAD software** — Dassault Systemes CATIA; Electronic design automation EDA software; OrCAD Capture; SpectraQuest; 1 more
- **Configuration management software** — Chef; HyperSpace; Perforce Helix software; Puppet; 8 more
- **Content workflow software** — Atlassian JIRA
- **Customer relationship management CRM software** — Blackbaud The Raiser's Edge; Oracle Eloqua; Salesforce software
- **Data base management system software** — Amazon DynamoDB; Apache Hive; Oracle Database; Oracle PL/SQL; 11 more
- **Data base reporting software** — Microsoft SQL Server Reporting Services SSRS; Oracle Business Intelligence Suite; SAP Crystal Reports
- **Data base user interface and query software** — Blackboard software; IBM DB2; ServiceNow; Transact-SQL; 11 more
- **Data mining software** — Google Analytics
- **Desktop communications software** — Remote control software; Skype; Stac Software ReachOut; Symantec pcAnywhere; 1 more
- **Desktop publishing software** — Microsoft Publisher
- **Development environment software** — Apache Kafka; Apache Maven; Oracle Java 2 Platform Enterprise Edition J2EE; Oracle SQL Developer; 25 more

- **Document management software** — Adobe Acrobat; Microsoft SharePoint
- **Electronic mail software** — IBM Notes; Microsoft Exchange; Microsoft Outlook
- **Enterprise application integration software** — Extensible markup language XML; IBM InfoSphere DataStage; Microsoft SQL Server Integration Services SSIS; WebFOCUS; 4 more
- **Enterprise resource planning ERP software** — Microsoft Dynamics; Oracle PeopleSoft; SAP ERP; SAP software; 7 more
- **Enterprise system management software** — IBM Power Systems software
- **Expert system software** — Ansible software
- **File versioning software** — Apache Subversion SVN; Git; Version control software
- **Financial analysis software** — Cost estimating software; Delphi Technology; Oracle E-Business Suite Financials
- **Geographic information system** — ESRI ArcGIS software; Geographic information system GIS software
- **Graphical user interface development software** — Salesforce Visualforce
- **Graphics or photo imaging software** — Adobe Illustrator; Adobe Photoshop
- **Helpdesk or call center software** — Help desk software
- **Human resources software** — ADP Workforce Now; Human resource management software HRMS; Oracle Taleo
- **Industrial control software** — Supervisory control and data acquisition SCADA software
- **Information retrieval or search software** — LexisNexis
- **Instant messaging software** — Blink
- **Internet directory services software** — Active directory software
- **Internet protocol IP multimedia subsystem software** — Voice over internet protocol VoIP system software
- **Medical software** — Epic Systems; Medical condition coding software; Medical procedure coding software; MEDITECH software; 1 more
- **Metadata management software** — Informatica Corporation PowerCenter; Oracle Master Data Management MDM Suite; Quest Erwin Data Modeler; SAP Master Data Management MDM
- **Network conferencing software** — Slido interaction software
- **Network monitoring software** — Nagios; Network intrusion prevention systems NIPS; Snort; Wireshark
- **Network security or virtual private network VPN management software** — Virtual private networking VPN software
- **Object or component oriented development software** — C#; jQuery; Scala; Swift; 17 more
- **Object oriented data base management software** — Hibernate ORM; Microsoft Visual FoxPro; PostgreSQL
- **Office suite software** — Microsoft Office software
- **Operating system software** — Bash; Microsoft Windows Server; Red Hat Enterprise Linux; UNIX Shell; 10 more
- **Pattern design software** — Diagramming software; Omni Group OmniGraffle
- **Portal server software** — Apache HTTP Server

- **Presentation software** — Microsoft PowerPoint
- **Process mapping and design software** — Flow chart software; Microsoft Visio
- **Program testing software** — Functional testing software; Hewlett Packard LoadRunner; JUnit ; Selenium ; 20 more
- **Project management software** — Atlassian Confluence ; Microsoft Team Foundation Server ; Microsoft Teams ; Oracle Primavera Enterprise Project Portfolio Management ; 1 more
- **Requirements analysis and system architecture software** — Architecture description language ADL; Popkin System Architect; Requirements management software; Unified modeling language UML; 1 more
- **Sales and marketing software** — Marketo Marketing Automation
- **Spreadsheet software** — Microsoft Excel
- **Storage networking software** — Amazon Simple Storage Service S3
- **Transaction security and virus protection software** — McAfee; NortonLifeLock cybersecurity software; Virus scanning software
- **Transaction server software** — Customer information control system CICS; Microsoft Internet Information Services (IIS); Sun Microsystems Sun ONE; Web server software
- **Video conferencing software** — Cisco Webex
- **Video creation and editing software** — YouTube
- **Web page creation and editing software** — Adobe Dreamweaver
- **Web platform development software** — Django ; Google Angular ; React ; Spring Framework ; 23 more
- **Word processing software** — 3M Post-it App; Google Docs ; Microsoft Word

Hot Technologies are requirements most frequently included across all employer job postings.
See all 119 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 10 In Demand skills for this occupation.

Essential Skills

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Importance	Essential Skill
75	**Speaking** — Talking to others to convey information effectively.
72	**Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
69	**Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.

Importance	Essential Skill
69	**Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
66	**Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making.
60	**Monitoring** — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
60	**Writing** — Communicating effectively in writing as appropriate for the needs of the audience.
53	**Learning Strategies** — Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things.
50	**Mathematics** — Using mathematics to solve problems.
31	**Science** — Using scientific rules and methods to solve problems.

Transferable Skills

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Importance	Transferable Skill
69	**Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.
69	**Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.
66	**Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
60	**Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
56	**Programming** — Writing computer programs for various purposes.
53	**Coordination** — Adjusting actions in relation to others' actions.

Importance	Transferable Skill
53	**Social Perceptiveness** — Being aware of others' reactions and understanding why they react as they do.
53	**Time Management** — Managing one's own time and the time of others.
53	**Troubleshooting** — Determining causes of operating errors and deciding what to do about it.
50	**Operations Analysis** — Analyzing needs and product requirements to create a design.
50	**Operations Monitoring** — Watching gauges, dials, or other indicators to make sure a machine is working properly.
50	**Quality Control Analysis** — Conducting tests and inspections of products, services, or processes to evaluate quality or performance.
50	**Service Orientation** — Actively looking for ways to help people.
47	**Instructing** — Teaching others how to do something.
47	**Management of Personnel Resources** — Motivating, developing, and directing people as they work, identifying the best people for the job.
47	**Technology Design** — Generating or adapting equipment and technology to serve user needs.
44	**Persuasion** — Persuading others to change their minds or behavior.
38	**Negotiation** — Bringing others together and trying to reconcile differences.
38	**Operation and Control** — Controlling operations of equipment or systems.
31	**Equipment Selection** — Determining the kind of tools and equipment needed to do a job.
28	**Management of Financial Resources** — Determining how money will be spent to get the work done, and accounting for these expenditures.
25	**Management of Material Resources** — Obtaining and seeing to the appropriate use of equipment, facilities, and materials needed to do certain work.
22	**Equipment Maintenance** — Performing routine maintenance on equipment and determining when and what kind of maintenance is needed.

Importance	Transferable Skill
22	**Repairing** — Repairing machines or systems using the needed tools.
16	**Installation** — Installing equipment, machines, wiring, or programs to meet specifications.

Knowledge

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Importance	Knowledge
91	**Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
68	**Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.
62	**English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
56	**Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.
55	**Administration and Management** — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources modeling, leadership technique, production methods, and coordination of people and resources.
50	**Education and Training** — Knowledge of principles and methods for curriculum and training design, teaching and instruction for individuals and groups, and the measurement of training effects.
49	**Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.
43	**Engineering and Technology** — Knowledge of the practical application of engineering science and technology. This includes applying principles,

Importance	Knowledge
	techniques, procedures, and equipment to the design and production of various goods and services.
39	**Administrative** — Knowledge of administrative and office procedures and systems such as word processing, managing files and records, stenography and transcription, designing forms, and workplace terminology.
36	**Personnel and Human Resources** — Knowledge of principles and procedures for personnel recruitment, selection, training, compensation and benefits, labor relations and negotiation, and personnel information systems.
34	**Communications and Media** — Knowledge of media production, communication, and dissemination techniques and methods. This includes alternative ways to inform and entertain via written, oral, and visual media.
33	**Economics and Accounting** — Knowledge of economic and accounting principles and practices, the financial markets, banking, and the analysis and reporting of financial data.
32	**Law and Government** — Knowledge of laws, legal codes, court procedures, precedents, government regulations, executive orders, agency rules, and the democratic political process.
31	**Public Safety and Security** — Knowledge of relevant equipment, policies, procedures, and strategies to promote effective local, state, or national security operations for the protection of people, data, property, and institutions.
28	**Design** — Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models.
27	**Sales and Marketing** — Knowledge of principles and methods for showing, promoting, and selling products or services. This includes marketing strategy and tactics, product demonstration, sales techniques, and sales control systems.
18	**Production and Processing** — Knowledge of raw materials, production processes, quality control, costs, and other techniques for maximizing the effective manufacture and distribution of goods.
14	**Geography** — Knowledge of principles and methods for describing the features of land, sea, and air masses, including their physical characteristics, locations, interrelationships, and distribution of plant, animal, and human life.
14	**Mechanical** — Knowledge of machines and tools, including their designs, uses, repair, and maintenance.

Importance	Knowledge
11	**Psychology** — Knowledge of human behavior and performance; individual differences in ability, personality, and interests; learning and motivation; psychological research methods; and the assessment and treatment of behavioral and affective disorders.
8	**Sociology and Anthropology** — Knowledge of group behavior and dynamics, societal trends and influences, human migrations, ethnicity, cultures, and their history and origins.
8	**Transportation** — Knowledge of principles and methods for moving people or goods by air, rail, sea, or road, including the relative costs and benefits.
7	**Philosophy and Theology** — Knowledge of different philosophical systems and religions. This includes their basic principles, values, ethics, ways of thinking, customs, practices, and their impact on human culture.
7	**Physics** — Knowledge and prediction of physical principles, laws, their interrelationships, and applications to understanding fluid, material, and atmospheric dynamics, and mechanical, electrical, atomic and sub-atomic structures and processes.
6	**Therapy and Counseling** — Knowledge of principles, methods, and procedures for diagnosis, treatment, and rehabilitation of physical and mental dysfunctions, and for career counseling and guidance.
5	**Chemistry** — Knowledge of the chemical composition, structure, and properties of substances and of the chemical processes and transformations that they undergo. This includes uses of chemicals and their interactions, danger signs, production techniques, and disposal methods.
5	**History and Archeology** — Knowledge of historical events and their causes, indicators, and effects on civilizations and cultures.
5	**Medicine and Dentistry** — Knowledge of the information and techniques needed to diagnose and treat human injuries, diseases, and deformities. This includes symptoms, treatment alternatives, drug properties and interactions, and preventive health-care measures.
4	**Biology** — Knowledge of plant and animal organisms, their tissues, cells, functions, interdependencies, and interactions with each other and the environment.
3	**Building and Construction** — Knowledge of materials, methods, and the tools involved in the construction or repair of houses, buildings, or other structures

Importance	Knowledge
	such as highways and roads.
0	**Fine Arts** — Knowledge of the theory and techniques required to compose, produce, and perform works of music, dance, visual arts, drama, and sculpture.
0	**Food Production** — Knowledge of techniques and equipment for planting, growing, and harvesting food products (both plant and animal) for consumption, including storage/handling techniques.
0	**Foreign Language** — Knowledge of the structure and content of a foreign (non-English) language including the meaning and spelling of words, rules of composition and grammar, and pronunciation.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

- Associate's degree or other 2-year degree required for some jobs
- Bachelor's degree required for some jobs
- Some college, no degree required for some jobs

Worker Characteristics

Abilities

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Importance	Ability
78	**Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
75	**Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).
75	**Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.

Importance	Ability
75	**Oral Expression** — The ability to communicate information and ideas in speaking so others will understand.
75	**Written Comprehension** — The ability to read and understand information and ideas presented in writing.
72	**Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
72	**Near Vision** — The ability to see details at close range (within a few feet of the observer).
72	**Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.
66	**Speech Recognition** — The ability to identify and understand the speech of another person.
63	**Fluency of Ideas** — The ability to come up with a number of ideas about a topic (the number of ideas is important, not their quality, correctness, or creativity).
63	**Speech Clarity** — The ability to speak clearly so others can understand you.
60	**Written Expression** — The ability to communicate information and ideas in writing so others will understand.
56	**Category Flexibility** — The ability to generate or use different sets of rules for combining or grouping things in different ways.
53	**Flexibility of Closure** — The ability to identify or detect a known pattern (a figure, object, word, or sound) that is hidden in other distracting material.
53	**Visualization** — The ability to imagine how something will look after it is moved around or when its parts are moved or rearranged.
50	**Far Vision** — The ability to see details at a distance.
50	**Mathematical Reasoning** — The ability to choose the right mathematical methods or formulas to solve a problem.
50	**Number Facility** — The ability to add, subtract, multiply, or divide quickly and correctly.

Importance	Ability
50	**Originality** — The ability to come up with unusual or clever ideas about a given topic or situation, or to develop creative ways to solve a problem.
50	**Perceptual Speed** — The ability to quickly and accurately compare similarities and differences among sets of letters, numbers, objects, pictures, or patterns. The things to be compared may be presented at the same time or one after the other. This ability also includes comparing a presented object with a remembered object.
50	**Speed of Closure** — The ability to quickly make sense of, combine, and organize information into meaningful patterns.
47	**Selective Attention** — The ability to concentrate on a task over a period of time without being distracted.
41	**Finger Dexterity** — The ability to make precisely coordinated movements of the fingers of one or both hands to grasp, manipulate, or assemble very small objects.
41	**Time Sharing** — The ability to shift back and forth between two or more activities or sources of information (such as speech, sounds, touch, or other sources).
38	**Memorization** — The ability to remember information such as words, numbers, pictures, and procedures.
38	**Visual Color Discrimination** — The ability to match or detect differences between colors, including shades of color and brightness.
25	**Auditory Attention** — The ability to focus on a single source of sound in the presence of other distracting sounds.
25	**Control Precision** — The ability to quickly and repeatedly adjust the controls of a machine or a vehicle to exact positions.
25	**Hearing Sensitivity** — The ability to detect or tell the differences between sounds that vary in pitch and loudness.
22	**Arm-Hand Steadiness** — The ability to keep your hand and arm steady while moving your arm or while holding your arm and hand in one position.
22	**Manual Dexterity** — The ability to quickly move your hand, your hand together with your arm, or your two hands to grasp, manipulate, or assemble objects.

Importance	Ability
19	**Trunk Strength** — The ability to use your abdominal and lower back muscles to support part of the body repeatedly or continuously over time without "giving out" or fatiguing.
16	**Wrist-Finger Speed** — The ability to make fast, simple, repeated movements of the fingers, hands, and wrists.
13	**Depth Perception** — The ability to judge which of several objects is closer or farther away from you, or to judge the distance between you and an object.
10	**Extent Flexibility** — The ability to bend, stretch, twist, or reach with your body, arms, and/or legs.
6	**Dynamic Strength** — The ability to exert muscle force repeatedly or continuously over time. This involves muscular endurance and resistance to muscle fatigue.
6	**Reaction Time** — The ability to quickly respond (with the hand, finger, or foot) to a signal (sound, light, picture) when it appears.
3	**Multilimb Coordination** — The ability to coordinate two or more limbs (for example, two arms, two legs, or one leg and one arm) while sitting, standing, or lying down. It does not involve performing the activities while the whole body is in motion.
3	**Rate Control** — The ability to time your movements or the movement of a piece of equipment in anticipation of changes in the speed and/or direction of a moving object or scene.
0	**Dynamic Flexibility** — The ability to quickly and repeatedly bend, stretch, twist, or reach out with your body, arms, and/or legs.
0	**Explosive Strength** — The ability to use short bursts of muscle force to propel oneself (as in jumping or sprinting), or to throw an object.
0	**Glare Sensitivity** — The ability to see objects in the presence of a glare or bright lighting.
0	**Gross Body Coordination** — The ability to coordinate the movement of your arms, legs, and torso together when the whole body is in motion.
0	**Gross Body Equilibrium** — The ability to keep or regain your body balance or stay upright when in an unstable position.

Importance	Ability
0	**Night Vision** — The ability to see under low-light conditions.
0	**Peripheral Vision** — The ability to see objects or movement of objects to one's side when the eyes are looking ahead.
0	**Response Orientation** — The ability to choose quickly between two or more movements in response to two or more different signals (lights, sounds, pictures). It includes the speed with which the correct response is started with the hand, foot, or other body part.
0	**Sound Localization** — The ability to tell the direction from which a sound originated.
0	**Spatial Orientation** — The ability to know your location in relation to the environment or to know where other objects are in relation to you.
0	**Speed of Limb Movement** — The ability to quickly move the arms and legs.
0	**Stamina** — The ability to exert yourself physically over long periods of time without getting winded or out of breath.
0	**Static Strength** — The ability to exert maximum muscle force to lift, push, pull, or carry objects.

Career Interest Types

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Occupational Interest	Career Interest Type
90	**Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.
82	**Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.

Occupational Interest	Career Interest Type
38	**Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
21	**Enterprising** — Work involves managing, negotiating, marketing, or selling, typically in a business setting, or leading or advising people in political and legal situations. Enterprising occupations are often associated with business initiatives, sales, marketing/advertising, finance, management/administration, professional advising, public speaking, politics, or law.
16	**Artistic** — Work involves creating original visual artwork, performances, written works, food, or music for a variety of media, or applying artistic principles to the design of various objects and materials. Artistic occupations are often associated with visual arts, applied arts and design, performing arts, music, creative writing, media, or culinary art.
15	**Social** — Work involves helping, teaching, advising, assisting, or providing service to others. Social occupations are often associated with social, health care, personal service, teaching/education, or religious activities.

Specific Interest Areas

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Occupational Interest	Specific Interest Area
93	**Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
54	**Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.
31	**Engineering** — Work involves applying science and technology to the design, building, testing, and use of electrical and electronic components, mechanical devices and machines, automotive, marine, and aerospace equipment and vehicles, materials, or structures.

Occupational Interest	Specific Interest Area
31	**Management/Administration** — Work involves planning, directing, or coordinating the daily operations of an organization, and overseeing personnel, business, or administrative services.
26	**Office Work** — Work involves clerical and administrative tasks for an organization such as answering phones, bookkeeping, updating records, scheduling appointments or meetings, and editing of documents and files.
23	**Mechanics/Electronics** — Work involves maintaining and repairing of machinery, such as automotive engines, electronic equipment, computers, and communication systems.
22	**Accounting** — Work involves examining, analyzing, and interpreting accounting records for financial statements, or maintaining, auditing, or evaluating records and accounts.
20	**Business Initiatives** — Work involves creating strategic plans or taking actions to expand an organization's business or market share.
13	**Finance** — Work involves directing and managing financial activities, such as planning and investments for an individual or organization to generate income.
13	**Teaching/Education** — Work involves teaching through lectures, discussions, and demonstrations in one or more subjects.
11	**Human Resources** — Work involves recruiting, selecting, and training job applicants, maintaining employee personnel records, administering compensation and employee-benefit programs, and conducting employee performance reviews.
10	**Public Speaking** — Work involves speaking before an audience to educate, entertain, or influence the listeners.
9	**Physical Science** — Work involves studying and researching non-living objects, materials, and phenomena through scientific disciplines such as physics, chemistry, astronomy, meteorology, or geology.
9	**Professional Advising** — Work involves providing expert advice, coaching, and counseling on topics such as careers, educational opportunities, and educational and professional development.
8	**Personal Service** — Work involves providing personal services to individuals in need of non-medical assistance such as helping arrange or plan for travel, meals,

Occupational Interest	Specific Interest Area
	or events.
7	**Health Care Service** — Work involves providing health care services to individuals to include the diagnosis, prevention, or treatment of diseases and other forms of illness or impairment.
5	**Sales** — Work involves selling goods or services.
5	**Social Science** — Work involves studying and researching human behavior, society, politics, and social relationships through scientific disciplines such as psychology, sociology, economics, or political science.
4	**Life Science** — Work involves studying and researching living organisms through scientific disciplines such as biology, botany, zoology, microbiology, physiology, biochemistry, or zoology.
4	**Marketing/Advertising** — Work involves promoting or creating a public image for products, services, or organizations to attract interest, engagement, or sales.
4	**Medical Science** — Work involves studying and researching the causes, nature, effects, and treatments of disease or other forms of impairment or illness.
3	**Applied Arts and Design** — Work involves applying artistic principles to the design and creation of various objects and materials.
3	**Humanities** — Work involves studying and researching aspects of human history, culture, and language through disciplines such as history, philosophy, the language arts, or area/ethnic studies.
3	**Media** — Work involves producing, developing, writing, directing, or presenting information through print, radio, television, or digital media.
3	**Protective Service** — Work involves maintaining order and protecting life and property by conducting investigations and enforcing laws and regulations to prevent or solve crimes.
2	**Creative Writing** — Work involves developing creative written works, such as novels, scripts, song lyrics, poetry, stories, or advertisements.
2	**Law** — Work involves studying and applying legal knowledge.
2	**Politics** — Work involves engaging in political activities to influence government policy.

Occupational Interest	Specific Interest Area
2	**Social Service** — Work involves providing social services and assistance to improve the well-being and social and psychological functioning of children and adults.
2	**Visual Arts** — Work involves creating original visual artwork for a variety of media and purposes using various techniques and software.
0	**Agriculture** — Work involves planting, cultivating, and harvesting crops, plants, or trees, or raising farm animals. Work can also involve preparing plant and animal products for distribution to consumers, or protecting natural resources to maximize their use in farming.
0	**Animal Service** — Work involves feeding, grooming, exercising, or otherwise providing care to promote and maintain the well-being of pets and other animals in settings such as kennels, animal shelters, and zoos.
0	**Athletics** — Work involves training, practicing, or competing in athletic events, or supporting those participating in athletics through coaching, training, or officiating.
0	**Construction/Woodwork** — Work involves constructing, installing, or repairing structures and fixtures made of wood.
0	**Culinary Art** — Work involves preparing, cooking, and presenting food in a distinctive style usually associated with a specific culture or region.
0	**Music** — Work involves conducting, composing, or playing music or singing.
0	**Nature/Outdoors** — Work involves planning, restoring, and conserving natural areas and wildlife habitats.
0	**Performing Arts** — Work involves acting as a performer in dance, comedy, or dramatic productions, choreographing of dance performances, or producing of live or recorded entertainment.
0	**Physical/Manual Labor** — Work involves manually unloading and moving freight, stock, luggage, or other materials, or performing other physical labor.
0	**Religious Activities** — Work involves conducting religious worship, leading a church or other religious group, and offering moral guidance associated with beliefs and practices of a religious faith.

Occupational Interest	Specific Interest Area
0	**Transportation/Machine Operation** — Work involves driving vehicles, such as vans, trucks, trains, and buses to transport people or materials. Work can also involve operating heavy equipment, such as cranes, forklifts, bulldozers, and backhoes to move materials or earth.

Work Styles

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Impact	Work Style
86	**Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.
78	**Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
77	**Innovation** — A tendency to be inventive, to be imaginative, and to adopt new perspectives on ways to accomplish work.
72	**Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.
62	**Adaptability** — A tendency to be open to and comfortable with change, new experiences, or ideas at work.
59	**Integrity** — A tendency to be honest and ethical at work.
58	**Perseverance** — A tendency to exhibit determination and resolve to perform or complete tasks in the face of difficult circumstances or obstacles at work.
57	**Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
57	**Tolerance for Ambiguity** — A tendency to be comfortable with ambiguity and uncertainty at work.
54	**Initiative** — A tendency to be proactive and take on extra responsibilities and tasks that may fall outside of one's required work role.

Impact	Work Style
53	**Achievement Orientation** — A tendency to establish and maintain personally challenging work-related goals, set high work-related standards, and exert high effort toward meeting those goals and standards.
53	**Cooperation** — A tendency to be pleasant, helpful, and willing to assist others at work.
53	**Stress Tolerance** — A tendency to cope and function effectively in stressful situations at work.
48	**Self-Confidence** — A tendency to believe in one's work-related capabilities and ability to control one's work-related outcomes.
38	**Self-Control** — A tendency to remain calm and composed and to manage emotions effectively in response to criticism or difficult situations at work.
30	**Social Orientation** — A tendency to seek out, enjoy, and be energized by social interaction at work.
25	**Sincerity** — A tendency to be genuine and sincere in interactions with others at work, without concern for personal gain or self-interest.
21	**Empathy** — A tendency to show concern for others and be sensitive to others' needs and feelings at work.
21	**Leadership Orientation** — A tendency to lead, take charge, offer opinions, and provide direction at work.
18	**Humility** — A tendency to be modest and humble when interacting with others at work.
6	**Optimism** — A tendency to exhibit a positive attitude and positive emotions at work, even under difficult circumstances.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025) $50.89 hourly, $105,850 annual

State wages Select a State Go

Local wages ZIP Code Go

Employment (2025) 544,400 employees

Projected growth (2025-2035) Much faster than average (7% or higher)

Projected job openings (2025-2035) 32,900

State trends Select a State Go

Top industries (2025)
Professional, Scientific, and Technical Services (33% employed in this sector)
Finance and Insurance (13%)
Management of Companies and Enterprises (10%)

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings Select a State Go

Local job openings ZIP Code Go

More Information

Related Occupations

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- 11-3021.00 Computer and Information Systems Managers Bright Outlook
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1242.00 Database Administrators
- 15-1243.00 Database Architects
- 15-1211.01 Health Informatics Specialists
- 15-1212.00 Information Security Analysts
- 15-1299.05 Information Security Engineers
- 15-1244.00 Network and Computer Systems Administrators
- 15-1252.00 Software Developers
- 15-1253.00 Software Quality Assurance Analysts and Testers

Supplemental related occupations

- 15-1251.00 Computer Programmers
- 15-1241.00 Computer Network Architects
- 15-1231.00 Computer Network Support Specialists
- 15-1232.00 Computer User Support Specialists
- 15-1243.01 Data Warehousing Specialists
- 13-1111.00 Management Analysts
- 15-1299.07 Blockchain Engineers
- 15-1299.04 Penetration Testers
- 15-2051.00 Data Scientists
- 15-1299.03 Document Management Specialists

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

All 6 displayed

- AFCEA International
- Association for Computing Machinery
- Computing Research Association
- IEEE Computer Society
- Institute of Electrical and Electronics Engineers
- National Center for Women and Information Technology

Accreditation, Certification, & Unions

- CompTIA
- Cyber Degrees
- Institute for Certification of Computing Professionals
- International Institute of Business Analysis
- Project Management Institute